

Troubleshooting Methodology -- Guided Notes ANSWER KEY

Unit 4.1 | CompTIA Network+ N10-009 Obj. Core 2 5.1 | N10-008

For instructor use / distribute after students complete the notes.

Question 1

The CompTIA troubleshooting methodology has _____ steps. Step 1 is _____ the problem. Step 2 is _____ a theory of probable cause. When gathering information in Step 1, open-ended questions (e.g., 'What were you doing when this happened?') are preferred over closed questions because they _____.

Answer: Seven (7); identify; establish; provide more detail / gather more information.

Question 2

Step 3 is _____ the theory. If the theory is confirmed, proceed to Step 4. If the theory is _____, establish a new theory or escalate. Step 4 is establishing a/an _____ of action to resolve the problem and identifying the potential effects of the fix.

Answer: Test; not confirmed (disproven); plan.

Question 3

Step 5 is _____ the solution or escalating as necessary, then _____ full system functionality and -- if applicable -- implementing preventive measures. Step 6 is _____ findings, actions, and outcomes, which creates a reference for future repairs and protects the technician legally.

Answer: Implementing; verifying; documenting.

Question 4

Step 7 is the final step: _____ the user. This includes explaining what caused the problem in non-technical terms, showing the user how to prevent the issue in the future, and confirming _____ with the resolution. This step is often skipped by junior technicians but is important for _____ and professional reputation.

Answer: Educate; satisfaction; customer satisfaction (client relationship).

Question 5

The distinction between a _____ and a _____ is critical in troubleshooting. The first is what the user observes or reports (e.g., 'the screen is black'). The second is the underlying technical reason (e.g., 'the GPU is not seated'). Treating a symptom without finding the _____ will result in the problem recurring.

Answer: Symptom; cause (root cause); root cause.

Question 6

Escalation means passing the problem to a _____ skill level or different team when it is beyond the current technician's ability to resolve. Before escalating, the technician should document what has

already been _____ and what the _____ confirmed or ruled out through testing.

Answer: Higher; tried (tested); tests (theory).

Question 7

When should a technician consider whether a _____ change occurred recently? If a user says 'it just stopped working,' a technician should ask about any _____, _____, or _____ changes in the recent past, as these are the most common causes of sudden failures.

Answer: Configuration; software updates; hardware changes; settings changes (any combination of recent changes).

Question 8 -- Real World Application

REAL WORLD: A user calls and says, 'My computer doesn't work.' Using the seven-step troubleshooting methodology, describe the exact questions you would ask in Step 1 and how those questions lead to a testable theory in Step 2. Demonstrate the difference between asking open-ended and closed questions.

Answer: Step 1 -- Identify the problem. Open-ended questions: 'What happens when you turn it on?', 'What were you doing when it stopped working?', 'Has anything changed recently -- updates, new hardware, dropped it?' These gather narrative detail. Closed questions (less useful in Step 1): 'Is it on?' -- yes/no only, provides minimal information. Step 2 -- Based on answers, form a theory. If the user says 'the screen is black but the fans spin,' the theory shifts from will not turn on to a display or POST failure. If the user says 'it happened after a Windows update,' the theory is software-related. Open-ended questions produce a richer symptom description, leading to a more targeted, testable theory.